THE VIRTUAL MEET EXPERIENCE

2017 - 2018

HS VIRTUAL CHALLENGE REGIONAL QUALIFIERS' MEET

COMPUTER SCIENCE

DO NOT OPEN TEST UNTIL TOLD TO DO SO

2017-2018 Virtual Challenge Regional Qualifiers' Meet – Computer Science

Note: Correct responses are based on Java, **J2sdk v 1.7.25**, from Sun Microsystems, Inc. All provided code segments are intended to be syntactically correct, unless otherwise stated (i. e. `error` is an answer choice) and any necessary Java 2 Standard Packages have been imported. Ignore any typographical errors and assume any undefined variables are defined as used. **For all output statements, assume that the System class has been statically imported…** ***import static java.lang.System.*;***

QUESTION 1

What is $77_8 + 113_{16}$?

A. 101010010_2 B. 338_{10} C. 522_8 D. 152_{16} E. All are equivalent

QUESTION 2

What is output by the code to the right?

A. `-18` B. `-9`

C. `0` D. `-25`

E. There is no output due to an error.

```
out.println(23%4-8/2*5+1*4-(6+6));
```

QUESTION 3

What is output by the code to the right ?

A. `\t1\n2\n34\t\t5\t6\n`

B.
```
        1        2        34        5  6
```

C.
```
1
2
3456
```

D.
```
        1
2
34        5      6
```

E. There is no output due to an error.

```
out.print("\t1\n2");
out.print("\n34\t\t");
out.print("5\t6\n");
```

QUESTION 4

What is output by the code to the right?

A. `nou`

B. `oRo`

C. `noug`

D. `oRou`

E. There is no output due to a syntax error.

```
String s="Baton Rouge";
String t=s.charAt(3);
t=t.concat(s.substring(7,9));
System.out.println(t);
```

QUESTION 5

What is output by the code to the right?

A. `true` B. `false`

```
boolean a=true || false && true;
boolean b= false || a || true;
boolean c=a && b ^ true;
boolean d=a ^ b ^ c;
out.print(a && c || b ^ d);
```

QUESTION 6

What is output by the code to the right?

A. `13.0` B. `13`

C. `17.0` D. `17`

E. There is no output due to an error.

```
int a=5,b=12;
out.println(Math.hypot(a, b));
```

QUESTION 7

What is the output by the code to the right?

A. `8.0`
B. `5.6`
C. `4.0`
D. `9.6`
E. There is no output due to an error.

```
float f=5.6;
double g=4;
g+=f;
out.print(g);
```

QUESTION 8

What is the output by the code to the right?

A. `north`
B. `northKorea`
C. `china`
D. `south`
E. There is no output due to an error.

```
String count="Korea";
boolean host=true;
if(count.equalsIgnoreCase("KoReA")) {
      if(host)
        out.print("north");
      else
        out.print("south");
    }
    else
      out.print("china");
      out.print(count);
```

QUESTION 9

What is the output by the code to the right ?

A. %%%%%%%%%%
B. %%%%%%%%%
C. %%%%%%%
D. %%%%%%%%
E. There is no output due to a syntax error.

```
int y=8;
do {
 out.print("%");
}while(--y>0);
out.print("%");
```

QUESTION 10

What is the output by the code to the right ?

A. `[4, 6, 4, 5, 6, 12, 0]`
B. `[2, 3, 4, 5, 6, 0, 1]`
C. `[6, 4, 5, 6, 12, 1, 0]`
D. There is no output due to a runtime error
E. Output cannot be determined until runtime.

```
int y[]= {2,3,4,5,6,1,0};
y[y[y[3]]]=y[y[2]];
y[5]=y[y[y[3]]]+y[4];
y[y[6]]=y[y[y[6]]];
out.print(Arrays.toString(y));
```

QUESTION 11

Which of the following could read in the following data?

```
212
```

A. `file.nextInt()`
B. `file.nextDouble()`
C. `file.nextLine()`
D. All of the above.
E. None of the above.

QUESTION 12

What is the runtime of the code to the right in Big O notation ?

A. `O(n*n)`
B. `O(logn)`
C. `O(n)`
D. `O(n*logn)`
E. There is no output due to a syntax error.

```
int h=0,n=12;
for(int y=0;y<n*2;y++) {
 for(int k=1;k<n;k*=2)
    h++;
}
```

QUESTION 13

Which of the operations to the right has the lowest precedence?

A. `III`
B. `IV`
C. `I`
D. `II`
E. `V`

```
I. ~
II. []
III. ++
IV. -
V. *
```

QUESTION 14

What is the output by the code to the right ?

A. `4`

B. `8`

C. `2`

D. `1`

E. `32`

```
out.println(Integer.BYTES);
```

QUESTION 15

What is the output by line marked `//1` in the code to the right ?

A. `true`

B. `false`

C. `8`

D. There is no output due to a syntax error.

E. There is no output due to a runtime error.

QUESTION 16

Assuming line `//1` works. what is the output by the line marked `//2` in the code to the right ?

A. `true`

B. `false`

C. `ksikknull`

D. There is no output due to a syntax error.

E. There is no output due to a runtime error.

```
ArrayList<Object>a;
a=new ArrayList<Object>();
a.add(new Object());
a.add(8);
a.addAll(a);
a.add(null);
a.add("ksikk");

out.println(a.get(0)==a.get(2)); //1

out.println
 (a.get(5) instanceof  a.get(4)); //2
```

QUESTION 17

What is the output by the code to the right ?

A. `39`

B. `30`

C. `33`

D. `36`

E. There is no output due to an error.

```
int y=3;
for(int h=0;h<12;h++)
 for(int g=0;g<h/2;g++)
    y++;
System.out.println(y);
```

QUESTION 18

What is output by the line marked `//1` code to the right?

A. `true 2`

B. `false 2`

C. `true 4`

D. `false 4`

E. There is no output due to an error.

QUESTION 19

What is output by the line marked `//2` code to the right?

A. `[212, 81]`

B. `[81, 212]`

C. Output cannot be determined until runtime.

D. There is no output due to a runtime error.

E. There is no output due to a syntax error.

```
HashSet<Integer>h,n;
h=new HashSet<Integer>();
n=new HashSet<Integer>();
h.add(212);h.add(8);h.add(4);
h.add(212);h.add(81);h.add(4);
n.add(12);n.add(8);n.add(4);
out.println
    (h.removeAll(n)+" "+h.size()); //1
out.println(h); //2
```

QUESTION 20

What is the big O runtime of accessing the first value in a doubly-linked list?

A. O(n)

B.O(n^2)

C.O(log(n))

D.O(n log(n))

E. O(1)

QUESTION 21

Which sorting algorithm is implemented in the code to the right?

A. Selection sort

B. Insertion sort

C. Quick sort

D. Merge sort

E. Spaghetti sort

```
int[]arr={2,3,4,1,7,3,8,5,5,3,7,1,2,2,4};
for (int i = 0; i < arr.length - 1; i++)
{
  int index = i;
  for (int j=i + 1; j < arr.length; j++)
     if (arr[j] < arr[index])
         index = j;
  int smallerNumber = arr[index];
  arr[index] = arr[i];
  arr[i] = smallerNumber;
}
```

QUESTION 22

What is the output by the line marked `//q22` in theclient code to the right?

A. `true`

B. `7`

C. `false`

D. Output cannot be determined until runtime.

E. This is no output due to an error.

QUESTION 23

What is the output by the line marked `//q23` in the client code to the right?

A. `true`

B. `12`

C. `false`

D. Output cannot be determined until runtime.

E. This is no output due to an error.

QUESTION 24

If the commented line in method solve is uncommented, what is the output by the line marked `//q22` in the client code to the right?

A. `[true, false, false, true, false, true, true, false]false`

B. `[true, false, false, true, false, true, true, false]true`

C. `[true, false, false, true, false, true, false, false, true, false]false`

D. `[true, false, false, true, false, true, false, false, true, false]true`

E. This is no output due to an error.

```
public static boolean
           solve(int[]stuff,int num)
{
   boolean[]fun=new boolean[num+1];
   fun[0]=true;
   for(int y=0;y<=num;y++) {
    if(fun[y]) {
      for(int i:stuff) {
        if(i+y<=num)fun[i+y]=true;
      }
    }
   }
   //out.print(Arrays.toString(fun));
   return fun[num];
}


//////////////////////////////////////
/////client code//////////////////////
out.println
 (solve(new int[] {8,5,3},7));  //q22
out.println(solve
  (new int[] {2,5,10,3},12));   //q23
```

QUESTION 25

What is the output by the code to the right?

A. `false 6 somebo`

B. `true 6 somebo`

C. `true 5 o`

D. `false 5 o`

E. This is no output due to an error.

```
String k=
"212 wildcats hurt somebody out there!";
out.print(k.matches
   ("\\d{2,5}(\\s[a-z]{2,11})+!")+" ");
String[]s=k.split("(\\w{2} )");
out.println(s.length+" "+s[3]);
```

QUESTION 26

What is output by the code to the right?

A. `{1, 2, 3, 4, 5, 6}`

B. `[1, 2, 3, 4, 5, 6]`

C. `1, 2, 3, 4, 5, 6`

D. `123456`

E. Output cannot be determined until runtime.

```
int[]d= {1,2,3,4,5,6};
out.print(Arrays.toString(d));
```

QUESTION 27

Which of the following is a valid java identifier?

A. `0stars`

B. `public`

C. `212`

D. `(*_*)`

E. All of the above are invalid.

QUESTION 28

What is output by the line marked `//q28`?

A. `66`

B. `69`

C. `67`

D. `70`

E. `68`

QUESTION 29

What is output by the line marked `//q29`?

A. `1`

B. `0`

C. `2`

D. `4`

E. `3`

QUESTION 30

What is output by the line marked `//q30`?

A. `61`

B. `62`

C. `63`

D. `64`

E. `60`

```
public static int recur(int i) {
  if(i>=9)
     return 1;
  return 3*i+recur(i+1)+recur(9);
 }

//////////////////////////////////////
/////////client code/////////////////
out.println(recur(6));  //q28
out.println(recur(212));  //q29
out.println(recur(-6));  //q30
```

```
class Structure<E>{ //question 31-34 refer to the following class
  int top;
  E[] con;
  int[]st;
  public Structure(int n) {
          con  =   <1*> ;
          st=new int[n];
  }
  public void add(int i,E e) {
          if(top==con.length)
            resize(top*2);
          con[top]=e;
          st[top++]=i;
  }
  public E get(int y) {
          return con[y];
  }
  public E remove() {
          E e=con[--top];
          con[top]=null;
          st[top]=-1;
          return e;
  }
  private void resize(int n) {
          E[] fun=(E[])(new Object[n]);
          int[]morfun=new int[n];
          System.arraycopy(con, 0, fun, 0, con.length);
          System.arraycopy(st, 0, morfun, 0, con.length);
          con=fun;
          st=morfun;
  }
}
/////client code//////////////////////////////////////////////////////////////////////
Structure<String>s=new Structure<String>(3);
s.add(2, "2");
s.add(3, "3");
s.add(5, "5");
s.add(1, "1");
s.add(7, "7");
s.add(6, "S");
System.out.println(s.remove()+" "+s.remove()); //1
System.out.println(Arrays.toString(s.st)+" "+s.top); //2
```

QUESTION 31

What could replace **<1*>** in the code to the right so that the `Structure` class works as intended?

A. `new E[n]`

B. `new Object[n]`

C. `(E[])(new Object[n])`

D. All of the above.

E. B and C

QUESTION 32

What is the output by the line in the client code marked `//1`?

A. `S 7`

B. `1 2`

C. `7 S`

D. There is no output due to an error.

E. `2 3`

QUESTION 33

What is the output by the line in the client code marked `//2`?

A. `[2, 3, 5, 1, -1, -1] 6`

B. `[-1, -1, 5, 1, 7, 6] 4`

C. `[-1, 3, 5, -1, 7, 6] 4`

D. `[2, 3, 5, 1, -1, -1] 4`

E. `[-1, 3, 5, -1, 7, 6] 6`

QUESTION 34

What data structure is implemented by the `Structure` class?

A. Map
B. Stack
C. List
D. Priority Queue
E. Queue

QUESTION 35

Which of the lines to the right first causes a syntax error?

A. `1`

B. `4`

C. `2`

D. `3`

E. There are no errors on the code to the right.

```
String k=3+""; //1
long g=1234L; //2
byte b=0b11111111; //3
short j=Float.MAX_VALUE; //4
```

QUESTION 36

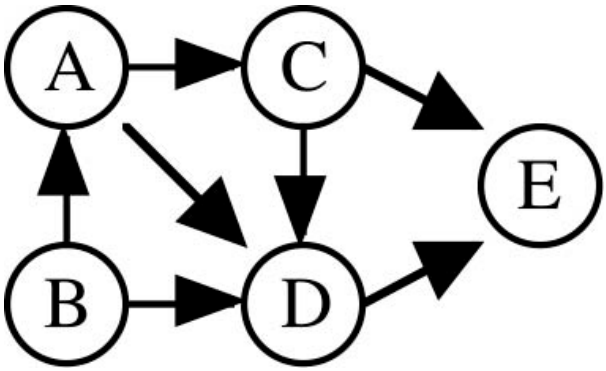
Which of the following is not an adjacent pair?

A. BA
B. AC
C. EA
D. CD
E. AD

QUESTION 37

Which of the folllowing describes the graph on the right?

A. `undirected unweighted`
B. `directed unweighted`
C. `undirected weighted`
D. `directed weighted`
E. `undirected unweighted and connected`



QUESTION 38

What is the signed binary 8-bit two's complement representation of the decimal number -45?

QUESTION 39

What is the infix equivalent of the following prefix expression (integer division)?

`4 5 % 4 3 2 7 17 12 * * * / + -`

QUESTION 40

What would a binary search tree look like if the following values were inserted(Level order)?

`H A B Y T W F E Q`

Name _____ Final Score _____ _____

School _____ 2nd Grading _____ _____

Division: 1A 2A 3A 4A 5A 6A Grade: 9 10 11 12 1st Grading _____ _____
score initials

The Virtual Challenge Meets
Computer Science - Student Answer Sheet

1)	_____	21)	_____
2)	_____	22)	_____
3)	_____	23)	_____
4)	_____	24)	_____
5)	_____	25)	_____
6)	_____	26)	_____
7)	_____	27)	_____
8)	_____	28)	_____
9)	_____	29)	_____
10)	_____	30)	_____
11)	_____	31)	_____
12)	_____	32)	_____
13)	_____	33)	_____
14)	_____	34)	_____
15)	_____	35)	_____
16)	_____	36)	_____
17)	_____	37)	_____
18)	_____	38)	_____
19)	_____	39)	_____
20)	_____	40)	_____

Total Number Correct _____ x 6 = _____

Total Number Incorrect _____ x -2 = _____

Final Score (record here and top right) _____

2017-2018 Virtual Challenge
Regional Qualifiers' Meet
Computer Science - KEY

1) E
2) D
3) D
4) E
5) A
6) A
7) E
8) B
9) B
10) A

11) D
12) D
13) B
14) A
15) A
16) D
17) C
18) A
19) C
20) E

21) A
22) C
23) A
24) A
25) B
26) B
27) E
28) C
29) A
30) A

31) C
32) A
33) D
34) B
35) D
36) C
37) B
38) 11010011
39) 4%5-(4+3/(2*7*17*12))
40) H A Y B T F W E Q

Note to Graders:

- All provided code segments are intended to be syntactically correct, unless otherwise stated (e.g. error is an answer). **Ignore any typographical errors**.
- Any necessary Standard Java 2 Packages are assumed to have been imported as needed.
- Assume any undefined (undeclared) variables have been defined as used.

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